Robredo, Joermine Elaine P. **Stat 143: Topic Proposal**
2011–03994 **2:30 – 4:00 PM**

Mental Health Literacy: A survey of a community's ability to recognize mental disorders and their beliefs about the effectiveness of treatment

Background of the Study

While public knowledge of physical diseases is mostly and widely accepted, knowledge about mental disorders has been comparatively neglected. As introduced and defined by Jorm et al (1997), 'mental health literacy' is the "*knowledge and beliefs about mental disorders which aid their recognition, management or prevention*." Mental health literacy is composed of "*(a) the ability to recognise specific disorders or different types of psychological distress; (b) knowledge and beliefs of risk factors; (c) knowledge and beliefs about self-help interventions; (d) knowledge and beliefs about professional help available; (e) attitudes which facilitate recognition and appropriate help-seeking; and (f) knowledge of how to seek mental health information*."

In the 2011, Mental Health Atlas from the World Health Organization, it was reported that there was a suicide rate of 2.5 per 100,000 population for males and 1.7 per 100,000 population for females. In the Philippines, neuropsychiatric disorders were estimated to contribute to 14.4% of the global burden of disease in a WHO 2008 report. The Philippine Statistical Authority reported that 1 in 5 adult Filipinos suffer from psychiatric disorders, ensuing an average of 88 reported cases of mental illness per 100,000 Filipinos.

Numerous studies indicate that there is a considerable impediment to the recognition and management of mental disorders. Recognition of the presence of a mental disorder may be high but only a minority may correctly give psychiatric label. Attitudes which hinder recognition and appropriate help-seeking are common.

Review of Related Literature

Mental health remains a neglected issue in most developing countries, especially in rural areas where access to effective mental health service is limited. Integration of mental health into primary healthcare is a way to direct this problem. Consequently, development of mental health awareness among communities is urgently needed. For effective results, it is important to understand and take account the local views on mental health and illness. Therefore, mental health literacy assessment has been done in rural areas to inform the development of a mental health training program (Kermode et al, 2009).

The community's knowledge and beliefs about mental health problems, their risk factors, treatments and sources of help may vary as a function of age. A survey was conducted during 2003-2004 with a national clustered sample of Australian adults aged 18 years and over. Case vignettes were presented describing depression and schizophrenia. Respondents were asked a series of question about their knowledge and beliefs. Age was coded into five categories and cross-tabulated with mental health variables. Assessment among age groups revealed that although older adults (70+ years) were poorer than younger age groups at correctly recognizing depression and schizophrenia, young adults (18-24 years) were more likely to misidentify schizophrenia as depression. Differences were also observed between younger and older age groups in terms of beliefs about the helpfulness of certain treating professionals and medical and lifestyle treatments for the mental disorders. (Farrer et al, 2008).

In a probability sampled population-based survey in Canada, the study's aim was to estimate and compare the percentages of personal stigma by levels of depression literacy and exposure to persons with depression, overall and by gender. A 9-item depression stigma scale was included in the questionnaire. The stigma variable was examined using a bivariate analyses and logistic regression modelling for each gender. In this study, more than 45% of the respondents considered that people with depression were unpredictable. Multivariate analyses on the association between case recognition, agreement with health professionals about treatment, personal contact with depression and stigma resulted a difference between gender (Wang and Lai, 2008). (Cotton, et. al, 2006) and (Gorczynski, et. al., 2016) also found that women revealed to have higher levels of mental health literacy than men.

Additionally, in another study, questions were asked about stigmatizing attitudes, including perceptions of discrimination, personal and perceived stigma and desire for social distance. Results showed that chronic schizophrenia was most likely to be related with dangerousness, unpredictability and a preference for not employing someone with the problem. Meanwhile, social phobia was most likely to be due to personal weakness. Compared to women, men with mental disorders had greater attitudes concerning dangerousness and social distance. Moreover, other people were more likely holding stigmatizing attitudes than the respondents stated for themselves. The study concluded that anti-stigma interventions were more likely to be effective if they aim on individual disorders rather than on mental illness in general (Reavley and Jorm, 2011).

The concept of mental health literacy assumes the superiority of expert psychiatric knowledge over lay beliefs. This could also be criticized for seeing the respondents' interpretation of their conditions as less valid (Jorm, 2000).

According to Tuliao, Filipino mental health help-seeking behavior is still limited, and mostly extrapolations from literature on Filipino Americans or Filipino Americans who studied alongside other Asian Americans (2014). A study on the influences of religious affiliation, religiosity, and spirituality upon help-seeking from religious clergy an mental health professionals were assessed after controlling for need, demographic, and cultural variables from 2,285 respondents for the Filipino American Community Epidemiological Survey. There was a comparable difference of 0.4% rate of help-seeking from mental health professionals from religious clergy. High religiosity was associated with more help-seeking from religious clergy but not less help-seeking from mental health professionals while high spirituality was associated with less mental health help-seeking (Abe-Kim et al, 2004).

Several authors (Burns and Rapee, 2006), (Kelly, Jorm, and Wright, 2007), (Goldney, Fisher and Wilson, 1999) similarly obtained that the most common source of recommended help was counselling. Although there is a positive attitude towards help-seeking, most people prefer to speak to friends or family than to seek help from professionals. However, in contrast, (Bland, Newman, and Orn, 1997) and (Andrews, Henderson and Hall, 2001) both found that there is a high proportion of those with a disorder who do not seek help.

Statement of the Problem

Improving mental health literacy is an important consideration when promoting convenient and effective treatment seeking for psychological disorders. Low recognition and stigma serve as a barrier to treatment. To assess the community's mental health literacy is a must to make a progress.

In the 2006 study, World Health Organization Assessment Instrument for Mental Health Systems was used to collect information on mental health system in the country. It aimed to assess the system and to provide mental health policy as there were no mental health legislation. It is also found in the study that there are mental health links with other relevant sectors, but there is no legislative or financial support for people with mental disorders. Non-standardized data are collected and compiled by facilities. However, studies done on mental health were not all published in indexed journals (WHO, 2007).

However, as mentioned, only a few studies done are published in indexed journals. Without much available information, one cannot be sure of how aware individuals of the different kind of mental disorders, how easily they recognize the signs and symptoms, and how they react when encountering one. The assessment is considerably needed to recommend a baseline mental health policy which will be easily understandable.

Definition of Variables

The following variables are needed for the study:

Age : age of the interview in years which will be classified (18-24, 25-39, 40-54, 55-69, 70+)

Gender : biological orientation of the respondent

Education : highest educational attainment

Religion : respondent's religious affiliation

Mental : knowledge on mental disorders (categorized as major depression ($Mental_1$) and schizophrenia ($Mental_2$))

Risk : knowledge on risk factors; measure by rating how likely each factor causes the disorder

Treatment : knowledge on the possible treatments; measure by rating each whether helpful, harmful, or neither

People : categorized as various people who can help mentally-ill patients; measure by rating each whether helpful, harmful, or neither

Stigma : personal attitude towards mental disorders; measure by stating how much the respondent agree to a statement

Health : respondent's current mental health condition; measure by using Kessler 10-item distress scale

Schematic Diagram

Based on the review of cited works, the proponent was able to devise the following diagram for the analysis of data. Demographic factors

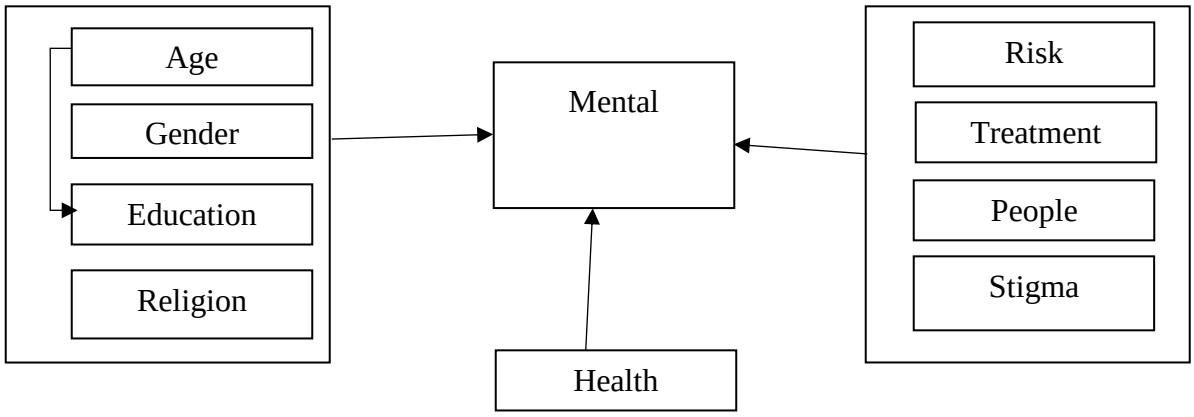


Figure 1. Schematic Diagram for the variables to be used

Limitations of the Study

The study is limited to the assessment of the implementation of the Mental Health Ordinance in Quezon City. This primarily aims to understand the mental health literacy in the barangays of Quezon City. Hence, knowledge and beliefs on mental disorders and the effectiveness of treatments will be the focus of the study which will be limited only with the selected barangays. Moreover, the study is also limited to the evaluation of the city health ordinance, the Community Based Mental Health Program of 2015.

Statistical tools to be used

1. Hypotheses

   By using a new case study that will be conducted in Quezon City, the study is designed to investigate how well the residents know about the mental health literacy. The researcher aims to check the association of knowledge to sex, age and educational attainment. Knowledge on various professionals and treatments on said subject will be examined. The following hypotheses for the literacy will be tested:

   H1: People easily recognize signs and symptoms of a person's mental illness.

   H2: Higher perception of mental health symptoms leads to easy classification of treatment from various professionals.

   H3: Demographic variables such as gender, age and educational attainment, influencing mental health literacy is contextual.

2. Method of data collection

   A two-stage sampling will be applied to the study which will be conducted in Quezon City. We treat barangays as cluster for a cluster sampling. To determine the number of cluster, we consider 141 barangays, based on the May 2010 statistics, to be randomly selected. Under each cluster, proportional simple random sampling without replacement will be conducted to determine the selected households. A list and map of numbered household in each selected barangay may be requested from concerned units, and households will be selected systematically from each barangay.

3. Statistical Procedures

   a. Descriptive Statistics

We measure the proportions for *gender*, *age*, and *education* variables on *Mental* variable.

`Table 1. Proportions of demographic variables with mental disorders.

		Sample	Depression		Schizophrenia	
		N (%)	Correct Symptoms	Wrong Symptoms	Correct Symptoms	Wrong Symptoms
Gender	Female					
	Male					
	p-value					
Age-group	18-24					
	25-39					
	40-54					
	55-69					
	70+					
	p-value					
Education	Primary School diploma					
	High School diploma					
	Bachelor's Degree					
	Master's Degree					
	Doctoral Degree					
	p-value					
Religion						
Knowledge about mental disorders						
Knowledge about various people who can help						
Knowledge about risk						
Stigma						
Health						

We measure independence and association among the variables *age, sex, and educational attainment* with variable *mental*.

Table 2. Association of demographic variables with mental disorders

Gender	Mental	
	Yes	No
Male		
Female		

We also include proportions of participants (1) providing various names for the conditions described, (2) identifying different actions to be taken for the people, (3) participants' perceptions of the helpfulness of various interventions to treat the conditions, and (4) helpfulness of various people who may be consulted by the people in the vignettes (based on open-ended questions).

Table 3. Proportions of respondents with classified responses

Recognized as…	Depression %	Schizophrenia %
Depression		
Nervous breakdown		
Schizophrenia		
Mental Illness		
Psychological problem		
Stress		
Has a problem		
Cancer		
Other		
Don't know		

b. Hypothesis Test
Analysis of data observation includes selection of variables, standardization of the data, multivariate analysis, indicator development and indicator indexing based on the way each of the variables influence mental health literacy.

c. Statistical Models
Analysis will involve chi-square test. All estimates will be compared per recognition of a mental health problem. Only differences significant at the 0.01 level ($P < 0.01$) will be reported (Jorm, et. al. 1997).

Data Collection Tool

A cross-sectional survey with structured interviews using vignettes of a person with either depression or schizophrenia will be used for a paper and pen interview with randomly selected respondents. The interview will be based on a vignette of a person suffering from a mental disorder. Half of the sample will be shown a vignette with a person suffering major depression and the other half would be with schizophrenia. The described person will be randomly assigned to a male or female. The interviewer will read out the case of the mentally-suffering person. The respondents will be asked two-open ended questions: (a) "What would you say if anything, is wrong with the person?" and (b) "How do you think the person could best be helped?"

Table 4. Case vignettes to be given/read to the respondents. (Farrer et. al, 2008)

Disorder	Vignette
Depression	John is 30 years old. He has been feeling unusually sad and miserable for the last few weeks. Even though he is tired all the time, he has trouble sleeping nearly every night. John doesn't feel like eating and has lost weight. He can't keep his mind on his work and puts off making decisions. Even day-to-day tasks seem too

	much for him. This has come to the attention of his boss, who is concerned about John's lowered productivity.
Early Schizophrenia	John is 24 and lives at home with his parents. He has had a few temporary jobs since finishing school but is now unemployed. Over the last six months he has stopped seeing his friends and has begun locking himself in his bedroom and refusing to eat with the family or to have a bath. His parents also hear him walking about his bedroom at night while they are in bed. Even though they know he is alone, they have heard him shouting and arguing as if someone else is there. When they try to encourage him to do more things, he whispers that he won't leave home because he is being spied upon by the neighbour. They realize he is not taking drugs because he never sees anyone or goes anywhere.

The rest of the questions for the interview is to determine the respondents' knowledge of and views on:

1. Various people who could help;
2. A range of possible treatments;
3. Knowledge of likely prognosis;
4. Knowledge of risk factors; and
5. Beliefs associated with stigma and discrimination

   Respondents will rate whether each sub-category are whether helpful, neither or harmful. Interviewers will undergo training to avoid bias and correct classification of the respondents' answers. The training includes an introduction to mental health and research methods, interviewing skills, sampling and recruitment and ethical aspects of research. All open-ended responses will be translated into English and will be systematically grouped thematically for quantification.

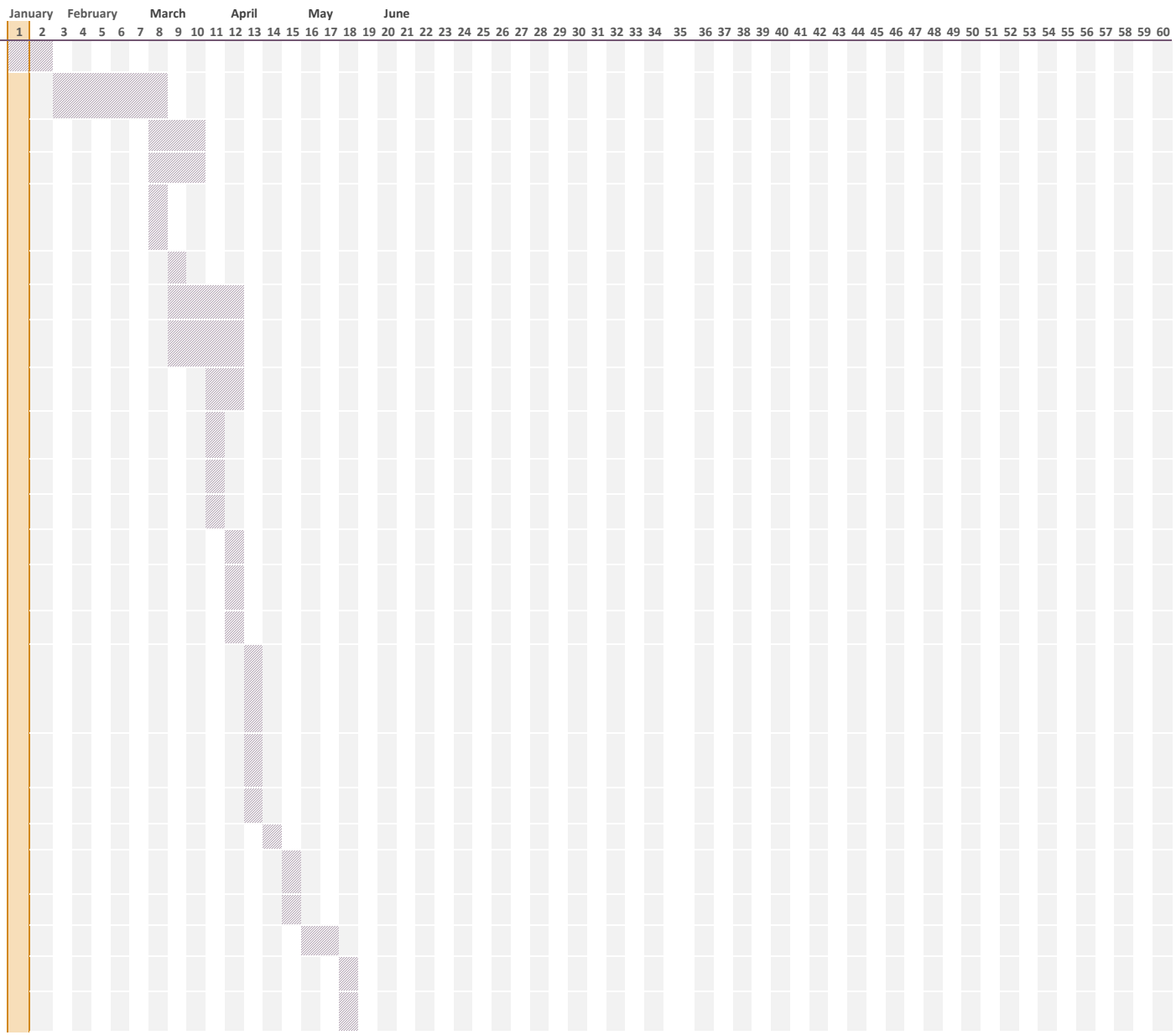
Mental Health Literacy: A Survey

1 Period = 1 Week.

Period Highlight 1

Plan Duration | Actual Start | % Complete | Actual (beyond plan) | % Complete (beyond plan)

ACTIVITY	PLAN START	PLAN DURATION	ACTUAL START	ACTUAL DURATION	PERCENT COMPLETE
Topic Proposal	1	2			
Planning and Design	3	6			
Questionnaire Draft	8	3			
Contact for funding	8	3			
Health Institutions, and Psychologists	8	1			
Contact Barangays	9	1			
Contact Trainers	9	4			
Look for Venue of Training	9	4			
Follow up Barangays	11	2			
Training of Interviwers	11	1			
Pretest 1	11	1			
Questionnaire Revision 1	11	1			
Pretest 2	12	1			
Questionnaire Revision 2	12	1			
Pilot Survey	12	1			
Distribution of flyers for announcement of survey	13	1			
Questionnaire Revision Final	13	1			
Actual Survey	13	1			
Encoding	14	1			
Editing/Cleaning Data	15	1			
Model Testing	15	1			
Drafting of Report	16	2			
Presentation	18	1			
Writing of Final Report	18	1			



Proposed Budget

PROJECTED CASH OUTFLOWS				
Items	Quantity	Unit	Cost/Item	Amount
Personnel:				
Office Staff (Salary)	4		5,000.00	60,000.00
Interviewers (training expenses)			6,000.00	6,000.00
Interviewers (Kits)	10		500.00	5,000.00
Trainers	2		10,000.00	20,000.00
Travel Fare			3,000.00	2,000.00
Office:				
Office Rental	1	room	12,000.00	36,000.00
Supplies (Printer)	2	pcs	4,000.00	8,000.00
Supplies (Paper)	10	reams	180.00	1,800.00
Supplies (Ballpen)	5	boxes	65.00	325.00
Supplies (Stapler)	10	pcs	35.00	350.00
Supplies (Folder)	10	sets	60.00	600.00
Internet + Telephone Bundle	2	unit	990.00	5,940.00
Training Venue Rental	1	room	3,000.00	3,000.00
Contingency Funds				14,901.50
Total				**163,916.50**

List of References

Abe-Kim, J., Gong, F., & Takeuchi, D. (2004). Religiosity, spirituality, and help-seeking among Filipino Americans: Religious clergy or mental health professionals? *Journal of Community Psychology*, 32(6), 675–689. doi:10.1002/jcop.20026

Bland, RC, Newman, SC and Orn H. Help-seeking for psychiatric disorder. *Can J Psychiatry*. Nov 1997; 42(9): 935-42. Date Accessed: 27 Feb 2017 <https://ncbi.nlm.nih.gov/pubmed>

Farrer et. al. Age differences in mental health literacy. *BioMed Central*. 20 April 2008. Date Accessed: 26 Feb 2017. <https://bmcpublichealth.biomedcentral.com>

Goldney, R., Laura Fisher and David Wilson. Mental health literacy: an impediment to the optimum treatment of major depression in the community. *Journal of Affective Disorders*. 29 Nov 1999. Date Accessed: 27 Feb 2017 <https://www.sciencedirect.com>

Jorm, A. et al. "Mental Health Literacy": a survey of a community's ability to recognise mental disorders and their beliefs about the effectiveness of treatment. *Medical Journal of Australia*. 1997. Date Accessed: 27 Jan 2017 <https://www.mja.com.au>

Jorm, A. F. Mental health literacy: Public knowledge and beliefs about mental disorders. *The British Journal of Psychiatry*. 2000. Date Accessed: 2 Feb 2017. <https://msrc.fsu.edu/ >

Kermode et. al. Community beliefs about treatments and outcomes of mental disorder: A mental health literacy survey in a rural area of Maharashtra, India. *Science Direct*. Public Health Vol 123, Issue 7. Jul 2009. Date Accessed: 5 Feb 2017 <http://www.sciencedirect.com/ >

K10 and K6 Scales. National Center for Health Statistics. *National Comorbidity Survey*. Date Accessed: 24 Feb 2017 <https://hcp.med.harvard.edu>

Mental Health Atlas 2011: Philippines. Department of Mental Health and Substance Abuse, World Health Organization. 2011. Date Accessed: 24 Feb 2017 <who.int/mental_health/evidence/atlas>

Paul Gorczynski, Wendy Sims-schouten, Denise Hill, Janet Clare Wilson, (2017) "Examining mental health literacy, help seeking behaviours, and mental health outcomes in UK university students", The Journal of Mental Health Training, Education and Practice, Vol. 12 Iss: 2, pp.111 - 120

Reavley, N. and Anthony F. Jorm. National Survey of Mental Health Literacy and Stigma. University of Melbourne. Dec 2011. Date Accessed: 27 Feb 2017 <http://pmhg.unimelb.edu.au>

Tuliao, Antover. Mental health help seeking among Filipinos: a review of the literature. *Taylor and Francis Online*. Date Published Online: 13 May 2014. Date Accessed: 7 Feb 2017 <http://www.tandfonline.com/ >

Wang, Jian Li and Daniel Lai. The relationship between mental health literacy, personal contacts and personal stigma against depression. *Journal of Affective Disorders*. September 2008. Date Accessed: 23 Feb 2017 <https://jad-journal.com>

WHO-AIMS Report on Mental Health System in the Philippines. World Health Organization. 2007. Date Accessed: 29 Jan 2017 <http://www.who.int>

APPENDIX A

--- --- --- --- --- Mental Health Literacy Questionnaire --- --- --- --- ---

Date: Name of Barangay:
Questionnaire No.: Interviewer:
Time Started:
Respondent Selection*: Member of the household between the ages of 18 and 74 years.*

Good day! My name is _____. I am a member of the research team from the School of Statistics, University of the Philippines Diliman. The group is conducting a survey on mental health literacy within Quezon City. The interview focuses on mental disorders and effectiveness of treatments. This survey aims to collect more information from residents like you about your knowledge and recognition of mental illnesses and beliefs of their treatments. This may take 15-20 minutes of your time.

Will you participate in the interview survey? All of your responses will be kept confidential as your name is not a required item in the questionnaire.

The answers you provide to the following questions will be of help to us in understanding more of the current scenario of mental health literacy. Please answer as accurately as possible. Thank you very much for your time.

Let's begin…
[Show assigned vignette to respondent.]

1. What would you say if anything is wrong with the person?

Possible Answers to describe the person in the vignette*
Depression
Nervous breakdown
Schizophrenia
Mental Illness
Psychological problem
Stress
Has a problem
Cancer
Shock
Trauma
Social problems
Other
Don't know
Refused

2. How do you think the person could best be helped?

Possible answers to be mentioned*
Talk over with friends and Family
See a doctor/General Physician
See a Psychiatrist

Take Medication
See a counsellor or have counselling
Person must first recognize the problem
Other
Don't know
Refused

*Use to classify respondent's answer.

3. Rate the various individuals that you think can help treat the disorder whether it is helpful, harmful or neither:

Help From…	**Helpful**	**Harmful**	**Neither**
Friends/Family			
General Physician			
Psychiatrist			
Medication			
Counselling			
Others			
Don't know			

4. Rate the various treatments that you think can help treat the disorder whether it is helpful, harmful or neither:

Treatment	**Helpful**	**Harmful**	**Neither**
General Physician			
Pharmacist			
Counsellor			
Social Worker			
Phone counselling			
Psychiatrist			
Psychologist			
Close family			
Close friends			
Naturopath/herbalist			
Clergy			
Vitamins, minerals			
Antidepressants			
Antibiotics			
Sleeping pills			
Antipsychotics			
Physical activity			
Read about problem			
Get out more			
Learn relaxation			
Cut out alcohol			
Psychotherapy			

Hypnosis			
Psychiatric ward			
Occasional drink			
Special diet			

5. Rate the likelihood of the following as causes for the disorder described in the vignette:

Likely Cause	Very likely	Likely	Not Likely
Virus or infection			
Allergies			
Day-to-day problems			
Recent death of a close friend or relative			
Experiencing a recent trauma			
Childhood problems			
Genetics			
Being a nervous person			
Weakness of character			

6. Rate the likelihood of the following as outcome for the person described in the vignette had appropriate professional help:

Likely Result	Very likely	Likely	Not likely
Full recovery with no further problems			
Full recovery, but problems would probably re-occur			
Partial recovery with no further problems			
Partial recovery, but problems would probably re-occur			
No improvement			
Get worse			
Don’t know			
Refused			

7. Rate the likelihood of the following as outcome for the person described in the vignette had **NO** appropriate professional help:

Likely Result	Very likely	Likely	Not likely
Full recovery with no further problems			
Full recovery, but problems would probably re-occur			
Partial recovery with no further problems			
Partial recovery, but problems would probably re-occur			
No improvement			
Get worse			
Don’t know			
Refused			

8. Rate following statements if you agree or not:

Statements	**Strongly Agree**	**Agree**	**Neutral**	**Disagree**	**Strongly Disagree**	**Not Applicable**
I would not vote for a politician with this problem.						
I would not employ someone with this problem.						
If I had this problem, I wouldn't tell anyone.						
People with this problem are unpredictable						
Avoid people with this problem.						
People with this problem are dangerous						
Problem is not a real medical illness						
Problem is a sign of personal weakness						
Person could snap out of the problem.						

9. Respondent's Information

a. Gender: o Male o Female

b. Age: ____

c. Educational attainment:

o primary school diploma
o high school diploma
o bachelor's degree
o master's degree
o doctoral degree

d. Civil Status: o Single o Married o Separated

e. Religion: _____

f. Rate your current health condition:

o Excellent o Good o Fair o Poor

10. Have you experienced any health problems for the past four weeks?

Health Problem	**Yes**	**No**
Colds		
Sore throat		
Headaches		
Dizziness		
Palpitations		
Breathlessness		

Backache		
Flu		
Anxiety		
Depression		
Tiredness		
Irritability		
Nervousness		

11. Use of Kessler 10 distress scale to identify respondent's score psychological distress (See Appendix B)

That ends the survey. Thank you very much.
Time Ended:

Sources: Jorm, 2000; Farrer et al, 2008; Reavley and Jorm, 2011.